

Under the Influence: A Survey of Large Language Models in Fake News Detection

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Abstract—Research into fake news detection has a long history, although it gained significant attention following the 2016 U.S. election. During this time, the widespread use of social media and the resulting increase in interpersonal communication led to the extensive spread of ambiguous and potentially misleading news. Traditional approaches, relying solely on pre-large language model (LLM) techniques and addressing the issue as a simple classification problem, have shown to be insufficient for improving detection accuracy. In this context, LLMs have become crucial, as their advanced architectures overcome the limitations of pre-LLM methods, which often fail to capture the subtleties of fake news. This literature review aims to shed light on the field of fake news detection by providing a brief historical overview, defining fake news, reviewing detection methods used before the advent of LLMs, and discussing the strengths and weaknesses of these models in an increasingly complex landscape. Furthermore, it will emphasize the importance of using multimodal datasets in the effort to detect fake news.

Impact statement—The imperative of detecting fake news holds considerable significance, bearing direct implications on human society. Automated detection of fake news remains a salient topic of discourse within scholarly communities. With the swift progression of artificial intelligence, we have transitioned into the epoch of LLMs. This review delineates the challenges and prospective directions associated with fake news detection in the context of these LLMs. The present article aims to give scholars a comprehensive overview of fake news detection during this era of LLMs, thereby aiding in streamlining subsequent inquiries and studies in the domain.

Index Terms—Fake news, fake news detection, large language models (LLMs), machine learning (ML).

I. INTRODUCTION

IN the ever-evolving landscape of media and technology, distinguishing between authentic information and fake news has emerged as a pressing challenge. The term ‘fake news,’ while not new, vaulted into the public lexicon following the

2016 U.S. elections; one exemplification of the power embedded in social media data and its capacity to propagate false information was witnessed during this time [1], [2]. Based on research conducted by [3], during the 2016 U.S. presidential election, approximately 6% of all news content on Twitter was identified as fake. However, the distribution was not uniform. A mere 1% of Twitter users encountered 80% of this fake news, and a fraction (0.1%) of users was accountable for disseminating 80% of it. This indicates a significant concentration of activity related to the dissemination and exposure to fake news [3]. Another research by [4] revealed traditional news sources performed better than fake news sources. However, the occurrence of fake news increased progressively over time. In these four studies [1], [3], [4], [5], originating from diverse fields such as economics, political science, and computer science, a consistent recommendation emerges: the need for a system to detect fake news automatically. The scope of inquiry extends beyond merely systems that autonomously identify false news; it also encompasses systems adept at offering an effective visual representation of the outcomes derived from the detection of fake news [6], [7]. However, what tangible results are associated with automatic fake news detection?

Based on several authoritative journals on fake news detection [8], [9], [10], [11], [12], it is evident that early research trends in this domain approached the issue as a classification problem, employing conventional methods like support vector machine (SVM), Naive Bayes, and logistic regression. Subsequently, the focus shifted towards incorporating natural language processing (NLP) techniques, such as word embedding for data preprocessing, and leveraging deep learning approaches, albeit within a classification framework. The issue today is that limited research has managed to enhance the precision of the existing state of the art by over 2% utilizing publicly accessible benchmark datasets [13]. This does not align with the rapid capabilities of large language models (LLMs) that can generate text swiftly [14].

LLMs, such as GPT-3, have brought new challenges and opportunities to the domain of fake news detection. These models, trained on vast amounts of data, can generate human-like text, making it increasingly difficult to distinguish between genuine and fabricated content. The sophistication of LLMs complicates traditional detection methods and necessitates the development of advanced techniques to identify AI-generated fake news. LLMs are significant in the fake news domain because they can understand and generate text with high contextual relevance and

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coherence. This ability allows them to create highly believable fake news, which traditional detection systems may struggle to identify. Moreover, LLMs can be used to enhance detection systems by providing better natural language understanding, enabling more accurate identification of misleading information. However, this dual capability of LLMs to both generate and detect fake news presents a complex challenge for researchers and practitioners in the field. Detecting fake news has increasingly become challenging [8], [15], [16], [17], [18]. How do LLMs impact this endeavor?

Consequently, our review paper is underpinned by three distinct aims.

- 1) To explain the concept of misinformation, commonly referred to as ‘fake news,’ which existed long before the internet, and to explore how it has evolved with the growth of digital platforms and the widespread use of social media in everyday life.
- 2) To explicate the dualistic nature of emergent technologies such as LLMs, which harbor the potential to serve as both a boon and a bane to society, depending on their application and governance.
- 3) To compile and assess the endeavors undertaken by the scholarly community to identify and combat fake news before the emergence of LLMs, to discuss the inherent limitations of fully automated fake news detection, and to delineate the prospective challenges that researchers face within this domain.

The organization of this article is outlined as follows: In Section I, we introduce the topic. Section II covers the methodology for selecting relevant papers. Section III defines and identifies fake news. The historical trajectory of fake news is discussed in Section IV. Traditional approaches to fake news detection before the rise of LLMs are discussed in Section V. Section VI discusses the datasets used in this field. The interplay between LLMs and fake news is explored in Section VII. Section VIII discusses how LLMs can be used to detect fake news, and Section IX presents advanced techniques in fake news detection. Section X offers a comparative analysis of detection methods before and after the advent of LLMs. Finally, the article concludes in Section XI and explores future directions in Section XII.

II. METHODOLOGY FOR SELECTING PAPERS

To ensure a comprehensive survey of the impact of LLMs on fake news detection, we utilized two primary data sources: Scopus and Google Scholar. These sources were selected due to their extensive coverage and reliability in indexing peer-reviewed research articles.

a) Scopus Search Strategy: An advanced search was performed in Scopus using a specific query (detailed in Appendix). This search strategy aimed to encompass a wide range of studies related to the impact of LLMs on fake news detection, highlighting the use of various datasets and methodological approaches.

b) Google Scholar and arXiv Supplementary Search: To ensure the inclusion of significant papers not indexed in Scopus,

supplementary searches were conducted on Google Scholar and arXiv using similar keywords. This step was essential to identify and incorporate substantial research contributions across different platforms and indexing systems.

III. UNDERSTANDING FAKE NEWS

In academic studies, fake news is typically described as information in the form of text, audio, or images that appears to be news but is actually false or deceptive. It is often crafted with the intention of misleading. Although different sources may provide slightly varied definitions, they agree on the key point: fake news involves a certain degree of deceit and distortion of the truth [1], [19], [20], [21], [22].

To fully understand fake news, it is essential to distinguish it from related terms such as misinformation, disinformation, and malinformation.

A. Misinformation

Misinformation refers to false or inaccurate information that is spread without the intent to deceive. It often results from misunderstandings, mistakes, or negligence. While misinformation can cause harm, it lacks the deliberate intent to mislead that characterizes fake news and disinformation [22].

B. Disinformation

Disinformation is intentionally false information created and disseminated with the explicit purpose of causing harm, misleading, or manipulating an audience. It is often used in the context of political propaganda, where the goal is to influence public perception or behavior [22].

C. Malinformation

Malinformation is information that is based on reality but used to inflict harm on a person, organization, or country. Unlike misinformation and disinformation, malinformation involves genuine information that is misused to create a damaging effect, such as through doxxing or leaking sensitive information [22].

In their extensive exploration of the term ‘fake news’ [20], [23], [24], [25], fake news can be differentiated into four specific categories, described below and summarized in Table I.

- 1) **Political propaganda:** False or misleading information disseminated for political gain. This could include inaccurate statistics, quotes taken out of context, or outright fabrications intended to manipulate public opinion [20], [23], [24], [25].
- 2) **Fabrication:** Completely made-up stories, often designed to generate online traffic and advertising revenue. This type of fake news is typically shared on social media and can spread rapidly [20], [23], [24], [25].
- 3) **Manipulation:** Real news that is manipulated in some way, for example, through misleading headlines or captions, or by selectively quoting from a source to distort the original meaning [20], [23], [24], [25].

TABLE I
EXAMPLES OF FAKE NEWS CATEGORIES AND TYPES

Category	Context	Example	Description	Type	References
Political propaganda	U.S.	“Pizzagate” conspiracy theory	False claims about democratic party officials operating a child sex trafficking ring.	Disinformation	[26]–[28]
	India	WhatsApp misinformation	Manipulated images or videos falsely depicting violent acts by certain religious or ethnic communities.	Disinformation	[29], [30]
	Brazil	2018 presidential elections	News articles making unverified claims about a candidate.	Disinformation	[31], [32]
	Russia	European elections interference	Fake news websites publishing false articles to tarnish the reputation of politicians.	Disinformation	[33], [34]
Fabrication	Fabrication	Breitbart’s report	A right-leaning news outlet suggested that Target’s stock prices had fallen due to its transgender policies, which lacks factual grounding.	Disinformation	[20], [35]
	Fabrication in scientific investigation	Andrew Wakefield’s work in 1998	Proposed a link between the MMR (measles, mumps, and rubella) vaccine and the occurrence of autism, which was later found to be fabricated.	Disinformation	[36]
Manipulation	Image data	Image Manipulation	Challenges of discerning between original and digitally manipulated images.	Disinformation	[37]
	Audio dataset	“Deep fake” audio clips	The rise of deep fakes, manipulated audio or video content created through AI.	Disinformation	[38], [39]
Parody or satire	News parody	The Onion	“Sexiest Man Alive” In 2012, an article was erroneously perceived as factual news, leading to its republication as genuine news.	Misinformation	[40]
	Political satire	The Borowitz report	“Congress Votes to Outsource Presidency” In 2013, an article was widely shared, with some readers treating it as actual news.	Misinformation	[41]
Malinformation	Real news misuse	Jennifer Lawrence photo leak	In 2014, private photos of Jennifer Lawrence were leaked online, which were true but meant to cause harm and embarrassment.	Malinformation	[27]

- 4) **Parody or Satire:** While not intended to deceive, this content can sometimes be taken out of context and shared as real news [20], [23], [24], [25].

An examination of the categories of fake news, as delineated in Table I, reveals a diverse range of contexts that spans across political propaganda, fabrication, manipulation, and even parody or satire, highlighting the variation of fake news context from scientific fabrications to erroneous celebrity news reporting. Furthermore, the dimensions of fake news extend beyond mere textual misinformation, encompassing audio and visual manipulations. As evidenced in Table I.

Having established the multifaceted nature of fake news, ranging from political misinformation to scientific fabrications and spanning across various media, including text, audio, and images (as outlined in Table I), it becomes vital to address the mechanisms for identifying and distinguishing these deceptive entities. In general, there are three primary mediums for the dissemination of fake news: text, images, and audio–video [42], [43]. Fig. 1 delineates the specific entities within these three mediums that can be utilized for fake news detection.

Synthesized from various references [42], [43], [44], [45], [46], the complexity and variety of fake news present an imperative for developing robust methods to discern truth from falsehood, regardless of the medium. Whether analyzing a controversial article, a digitally altered image, or a “deep fake” audio clip, the challenge lies in devising accurate, efficient, and adaptive solutions to recognize and counteract this pervasive phenomenon. The following Fig. 1 is created by summarizing from various sources [47], [48], [49].

As illustrated in Fig. 1, to investigate the dissemination of fake news across diverse media, it is imperative first to understand the information encapsulated within the news. In evaluating **text content**, source credibility and alignment with verified information are critical. Linguistic patterns, such as sensationalism and poor grammar, may indicate fake news, and a lack of source citation is suspicious. Verification can be enhanced by cross-checking with recognized fact-checking sites [50], [51]. For **audio content**, voice consistency and spectral analysis are crucial for authenticity [52]. In analyzing **images content**, digital tampering can be identified through inconsistencies in metadata and visual artifacts [53]. Tools like reverse image

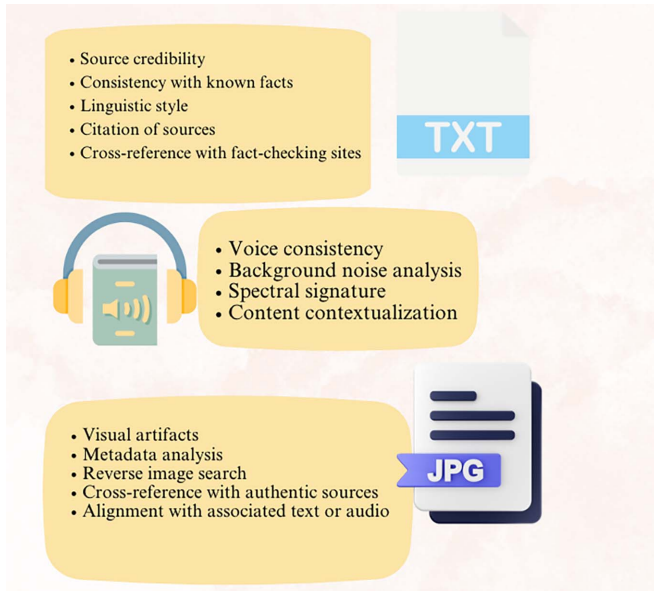


Fig. 1. Criteria-based analysis of fake news across mediums.

searches can validate image origins, ensuring consistency with associated content [54]. Adhering to these guidelines aids in distinguishing genuine from false content in the digital era.

IV. A BRIEF HISTORY OF FAKE NEWS

In this section, we will present a brief account of the history of fake news, categorized into three distinct timeframes: the pre-social media and Internet era, the pre-LLM or social media and Internet era, and the latest LLM era.

Before the rise of digital technology and social media, the propagation of fake news or misinformation was primarily facilitated through traditional media such as print, radio, television, and word-of-mouth. Notably, the “Great Moon Hoax” of 1835 is an illustrative case where the New York Sun published a series of entirely fictional articles about the discovery of life on the moon by a British astronomer [55]. The articles, which included elaborate descriptions of lunar forests, oceans, and bizarre creatures, were widely believed and reprinted. In another instance, the infamous “War of the Worlds” radio broadcast in 1938 caused widespread panic when listeners, missing the opening disclaimer, believed they were hearing actual news of a Martian invasion [56]. Television has also served as a platform for misinformation, as seen in the 1998 Operation Tailwind report by CNN and Time magazine, which falsely claimed the U.S. used nerve gas during a secret mission in the Vietnam War; the story was later retracted [57]. Additionally, word of mouth has proven a powerful medium for spreading misinformation, exemplified by the “Satanic Panic” in the 1980s and 90s, where stories of widespread satanic ritual abuse, lacking any substantial evidence, spread across the United States, causing considerable social distress [58]. These historical examples demonstrate how misinformation can infiltrate various media forms, underscoring the persistent need for media literacy and critical thinking.

Moving swiftly into the age of the Internet and social networks. The advent of the Internet and social media platforms revolutionized how information, including fake news, is disseminated. According to [59], platforms like Facebook, Twitter, and YouTube have democratized information dissemination, making it easy for individuals to share content with a broad audience. However, the convenience of sharing also opens up opportunities for spreading false information, creating the potential for misinformation campaigns. For instance, in the 2016 U.S. election [1], the adoption of social media data emerged as a pivotal component of the campaign blackprints for both Donald Trump and Hillary Clinton. Each campaign utilized targeted advertisement on platforms such as Facebook to access distinct demographics; significantly, the Trump campaign enlisted the services of Cambridge Analytica, a political data firm recognized for its microtargeting prowess employing data acquired from millions of Facebook profiles [60]. They exploited social media platforms to interact with voters, galvanize supporters, and broadcast campaign messages. Trump’s contingent was notably proactive on Twitter, adopting a provocative approach, whereas Clinton’s campaign adhered to a more conventional social media methodology. Furthermore, the data harvested from social media were scrutinized to measure public sentiment, pinpoint crucial issues, and fine-tune campaign strategies as needed. The 2016 election concurrently observed a substantial escalation in the dissemination of fake news across social media platforms. Misinformation was occasionally shared and circulated by adherents of both campaigns, often inadvertently. Allegations were made that fake news could have swayed voters’ perceptions and actions, with both campaigns being accused of disseminating or gaining from misinformation. However, the magnitude and repercussions of fake news are intricate to ascertain. The assertive and unconventional social media tactics of the Trump campaign are frequently acknowledged as a contributing element to his electoral triumph, facilitating direct engagement with the populace, circumventing traditional media outlets, and energizing his base [61]. Conversely, the Clinton campaign, notwithstanding a robust social media presence, was frequently perceived as more restrained and conventional in its methodology. The mechanics of sharing and the structure of social media networks often lead to the creation of echo chambers, where users predominantly receive and engage with information that aligns with their beliefs, further reinforcing misinformation [62]. The issue of anonymity on the Internet can make it challenging to verify the credibility of sources, which can be exploited to spread fake news without direct repercussions [27].

V. FAKE NEWS DETECTION BEFORE LLMs

In Section VIII, we elucidated the concept of LLMs concerning the issue of fake news. In the upcoming section, we will delve into the phenomenon of fake news by retracing our steps to a time preceding the advent of LLMs. Utilizing the Scopus database with search terms “fake news detection” AND “machine learning (ML),” we identified 156 relevant publications from 2016 to 2022. The overarching landscape of these

TABLE II
SUMMARY OF FAKE NEWS DATASETS. *FEATURE* → TEXT: ▼, IMAGES: ▼, VIDEOS: ●, SOCIAL CONNECTIONS: ●;
SOURCE → TWITTER: 🐦, FACEBOOK: 📘, OTHERS: 🌐; *ACCESSIBILITY* → GITHUB: 📄, PUBLICLY AVAILABLE: 📄; *LANGUAGE* →
ENGLISH = EN, CHINESE = ZH-CN

Dataset	Year	Source(s)	Accessibility	Features	Language(s)
MediaEval [67]	2015	🐦 📘 🌐	📄	▼ ▼ ●	en
PHEME [68] & PHEME Updated Version [69]	2016 & 2018	🐦	📄	▼ ●	en
Weibo [74]	2016	🌐	📄	▼ ▼ ●	en & zh-cn
Twitter15 [77] & Twitter16 [77]	2017 & 2017	🐦	📄	▼ ▼	en
Kaggle Fake News Dataset [78]	2018	🌐	📄	▼	en
FakeNewsNet [73]	2019	🐦	📄	▼ ●	en
COVID-19 Datasets [70], [71], [72]	2020–2021	🐦	📄	▼ ●	multi-language

4) *Logistic Regression*: It is a statistical method used for binary classification tasks. Unlike linear regression, which predicts continuous values, logistic regression predicts the probability of a given instance belonging to a particular category. The output is transformed using the logistic (or sigmoid) function to yield a value between 0 and 1. If the predicted probability exceeds a set threshold, typically 0.5, the instance is classified into one category; otherwise, it's classified into the other. The algorithm estimates the coefficients of the features using methods like maximum likelihood estimation to best fit the data. Logistic regression is widely used due to its simplicity and interpretability [8], [32], [63].

B. Deep Learning-Based Solutions

1) *Convolutional Neural Network (CNN)*: It is a deep learning algorithm crafted for the processing of structured grid data, predominantly focused on visual data like images and videos. It finds extensive utility in computer vision applications such as image categorization, object recognition, and facial identification. CNNs exhibit their distinctive ability to autonomously acquire hierarchical patterns and features from input data through the application of convolutional and pooling layers, rendering them exceptionally efficient in tasks related to visual content. Their pivotal role is evident in driving advancements in artificial intelligence domains, including autonomous navigation, medical image analysis, among others [14], [47] [64] [65].

2) *Recurrent Neural Network (RNN)*: It represents a category of artificial neural networks tailored for the analysis of sequential data, encompassing textual content, speech, and temporal data sequences. RNNs are endowed with an inherent memory mechanism, affording them the capability to factor in preceding information while handling present input, rendering them particularly apt for applications that demand consideration of contextual relationships and sequence chronology, exemplified by language modeling and speech recognition tasks [14], [47], [64], [65].

3) *LSTM*: It is a type of recurrent neural network (RNN) architecture. RNNs are designed to handle sequential data by maintaining a memory of previous inputs in their internal state. However, standard RNNs can struggle with long-term dependencies due to issues like vanishing or exploding gradients. LSTM addresses these challenges by introducing memory cells

and three gating mechanisms: input, forget, and output gates. These gates regulate the flow of information, allowing LSTMs to retain or forget data over longer sequences, making them particularly effective for tasks like time series forecasting, NLM, and speech recognition [14], [47], [64], [65].

4) *Bidirectional Encoder Representations From Transformers (BERT)*: It represents an advanced model in the field of NLM. BERT excels in comprehending contextual information within text by taking into account the words preceding and following a specific word. This bidirectional approach notably enhances its proficiency in comprehending and producing human-like language, rendering it a formidable resource for an array of language-centric applications, including but not limited to text classification, question answering, and other related tasks [46], [66].

VI. DATASETS

In this section, we elaborate on the datasets commonly utilized in fake news detection research, with additional details presented in Table II and Table III. The ‘Dataset’ refers to the specific name of the dataset employed. ‘Year’ denotes the year in which the dataset was published. The ‘Size’ of the dataset indicates the volume or the number of data points it encompasses. ‘Source(s)’ details the origin or platforms from where the dataset can be obtained. ‘Accessibility’ describes the means through which researchers and users can access the dataset, including any public availability, required permissions, or restrictions. ‘Features’ outline the types of content present in the dataset, such as text, videos, images, etc. Finally, ‘Language(s)’ specifies the language(s) in which the dataset’s content is available, providing a comprehensive understanding of the dataset’s characteristics and how it can be accessed in the context of fake news detection research.

A. MediaEval

The dataset was used in the “verifying multimedia use” competition in MediaEval2015 and MediaEval2016, a benchmarking event for multimedia algorithm evaluation. This dataset, primarily from Twitter, aids in building classifiers for multimedia posts. It’s notable for its abundance of posts with images and videos [67].

B. PHEME

It is a dataset with 330 threads on nine major events, like the Ottawa shooting and Ferguson unrest, sourced from Twitter. It categorizes tweets by story and verifies them as true, false, or unverified [68].

C. PHEME Updated Version

It is an expanded version, categorizing threads as rumors or nonrumors, then validating the rumors. It also annotates tweet stances through crowdsourcing. The dataset has 2402 rumors: 1067 true, 638 false, and 697 unverified [69].

D. COVID-19 Datasets

in this section we summarize several sources, including: **CoAID** is a dataset focused on COVID-19 related fake news detection from social media. It contains 4251 news articles and 296,000 user interactions, collected between 1 December 2019, and 1 September 2020 [70]. **COVID-HeRA** is an expanded version of the **CoAID** dataset, aims to identify unreliable posts by assessing the risk and impact of COVID-19 misinformation on health decisions. It is categorized into five groups: real news/claims, not severe, possibly severe misinformation, highly severe misinformation, and refutes/rebuts misinformation [71]. **ArCOV19-Rumors** features 162 authenticated claims and corresponding tweets. The labeling in ArCOV19-Rumors supports two misinformation detection types on Twitter: claim-level verification (classifying each claim and its related tweets as fake or not) and tweet-level verification (a two-class classification based on a tweet and its propagation, including replies and re-shares) [72].

E. FakeNewsNet

It is a widely-used dataset for fake news detection, offering a deep social media context. It features articles and associated tweets verified by PolitiFact or GossipCop. With 467 thousand tweets from PolitiFact and 1.25 million from GossipCop, they're labeled as real or fake. It provides rich contextual data, including user profiles and social networks. However, collecting tweets using the Twitter API can be time-consuming due to the data's volume [73].

F. Weibo

The dataset contains verified labels. The writers gathered all confirmed false rumor posts from May 2012 to January 2016 from Weibo's official rumor debunking system. This platform allows users to flag doubtful tweets, which a respected committee then reviews to determine their veracity [74].

G. Twitter15 & Twitter16

According to [75] and [76], these are the benchmark datasets. They collect highly retweeted source tweets and their associated propagation threads on Twitter. These source tweets are categorized into four labels based on their originating events: true rumors, false rumors, nonrumors, and unverified rumors [77].

Furthermore, we incorporate statistical data on the number of publications referencing papers associated with the dataset, as derived from Google Scholar, as depicted in Fig. 3.

The pie chart in Fig. 3 illustrates the prevalence of various benchmark datasets in fake news detection research, revealing significant insights into their usage within the academic community. The most utilized dataset, FakeNewsNet, dominates the chart, highlighting its critical role in the field. PHEME and its updated counterpart, along with Twitter15 and Twitter16, represent substantial portions, underscoring their importance for analyzing rumors and misinformation on social media. Weibo's notable presence indicates a focus on Chinese social media, while CoAID and MediEval show moderate usage, suggesting specialized applications in health misinformation and historical texts. Lesser-used datasets like COVID-HeRA and ArCOV19-Rumors point to niche research areas, particularly around COVID-19. Overall, the distribution underscores the diverse approaches in tackling fake news, with each dataset offering unique insights for different research focuses.

VII. THE RELATIONSHIPS BETWEEN LLMs AND FAKE NEWS CONTENT

Detecting fake news has become even more challenging in the present era of LLMs. These models, such as GPT-3, are sophisticated artificial intelligence systems trained on vast amounts of data to generate human-like text [90]. While they have many valuable applications, LLMs also pose risks in generating or amplifying misinformation [91]. Fake news generated by these models can be highly deceptive, as the models can produce contextually relevant and coherent responses. The challenge lies in distinguishing between genuine and fabricated information, as the linguistic patterns employed by these models can closely resemble authentic content [92]. Furthermore, the rapid dissemination of information through social media and the Internet can exacerbate the spread of fake news generated by LLMs [93]. The experiment carried out by [66] aimed to identify fake news by utilizing a dataset acquired from ChatGPT's creation named CheckThat-2022, marking the fifth cycle of the CheckThat! lab, a component of the 2022 Conference and Labs of the Evaluation Forum (CLEF). In this endeavour, the researcher employed mBERT² to differentiate among four labels: true, partially false, false, and other. Despite these efforts, the outcomes remained subpar, with an accuracy rate of 0.3 and an F1 score of 0.25 [66]. Efforts to address this challenge involve developing specialized techniques to detect AI-generated fake news and ensuring responsible deployment and use of LLMs.

A. LLMs as Content Generators

In deep learning and NLM, LLMs (LLMs) such as those in the GPT series represent a significant advancement [94]. Utilizing the Transformer architecture—a neural network structure adept at handling sequences—these models demonstrate remarkable proficiency in text processing. The training regimen

²(Multilingual BERT) is a pretrained language representation model designed to understand text across multiple languages.

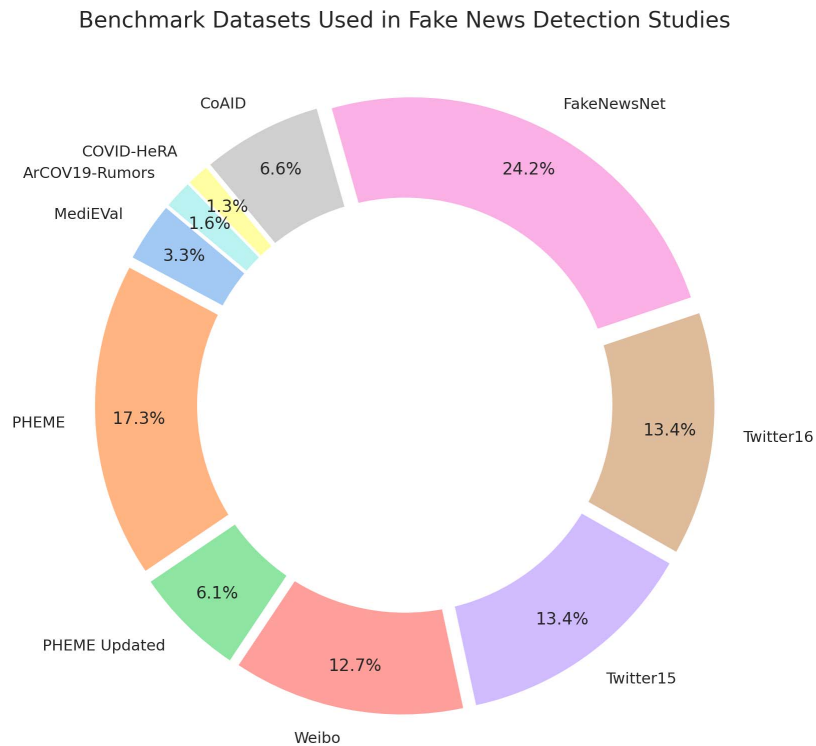


Fig. 3. Statistical overview of scholarly citations for dataset-related papers as indexed by Google Scholar.

for LLMs encompasses two distinct phases. The initial phase involves unsupervised pretraining on vast textual datasets, enabling the model to anticipate succeeding words in textual sequences, thereby acquiring a generalized linguistic understanding. After this, fine-tuning is performed on specialized tasks, enhancing the model’s adaptability. A pivotal feature within this architecture is the “attention mechanism,” which facilitates the model’s discernment in allocating weightage to words contingent on contextual relevance. Postcomprehensive training, these models possess the capability to produce text that is both contextually pertinent and coherent, stemming from user-defined prompts. This Transformer-centric approach can trace its origins to the foundational paper by [95]. Building upon the aforementioned prowess of LLMs in emulating human-like text generation capabilities, it is evident that the implications of such competencies are double-edged. While LLMs can construct narratives that parallel human compositions, an associated risk emerges. The potential for these models to be commandeered for crafting fake news or misleading narratives underscores a pressing concern in the digital age. The dilemma surrounding the authenticity of content spawned by LLMs and its potential misuse is not just a theoretical concern [96], [97], [98]. Study outlined by [99], utilizing ChatGPT in such a way could potentially foster the proliferation of fake news, as individuals might duplicate and disseminate information devoid of appropriate attribution, culminating in plagiarism [99].

Building upon the deliberation above on LLMs’ capabilities, analyzing their development trajectory becomes imperative. Across multiple iterations of LLMs, there has been a

marked enhancement in the quality, fluency, and authenticity of the generated synthetic text. Initial LLM versions, while showcasing essential linguistic correctness, occasionally grappled with coherence and contextual pertinence [100], [101], [102]. However, as these models expanded in architectural complexity and were exposed to more extensive training datasets, there was a noticeable elevation in their output’s human-like fluency and contextual depth [102]. By the advent of models akin to GPT-3, the distinctions between machine-generated and human-authored content became remarkably subtle, emphasizing both the marvels of AI advancements and the concomitant vigilance required [102], [103], [104], [105]. For example, we generated news about a public figure, Kris Jenner, from ChatGPT, utilizing the keyword, kindly produce news regarding Kris Jenner.

In her announcement, Jenner stated, “Jenner Elegance is a true passion project for me. It’s all about bringing luxury, elegance, and style into everyday life.” She promises the brand will deliver high-quality, stylish products that reflect her style.

Contrary to what has been circulating, Kris Jenner has never owned a jewelry line called ‘Jenner Elegance’. Thus, this information can be categorized as misinformation or ‘fake news’. LLMs are recognized for their capability to craft texts that are not only coherent but also contextually pertinent, often mirroring the quality of human-authored content [102], [106]. Analogous to a search engine’s autocomplete feature, LLMs have the proficiency to reorganize and repurpose the data they’ve learned from. Notably, while there might be occasional errors, these models exhibit their text synthesis prowess by generating

TABLE III
DATASET STATISTICS AND CHARACTERISTICS

Dataset	Details
MediaEval [67]	Flickr queries for development: 153 (one-concept); images for development: 45 375; Flickr queries for testing: 139 (69 one-concept, 70 multiconcept); images for testing: 41 394; metadata, Wikipedia pages, and content descriptors for text and visual modalities; data annotated for the relevance and diversity of photos.
PHEME [68]	Users: 56 099; rumor source tweets: 208; users responded to rumors: 3129; retweets of rumors: 53 469; judgments on tweets: 68 247; annotated threads: 330; tweets: 4842; inter-annotator agreement: 62.3%
PHEME updated version [69]	Threads: 6425; tweets: 105 354; rumors: 2402; non-rumors: 4023; true rumors: 1067; false rumors: 638; unverified rumors: 697
Weibo [74]	Rumor (training set): 3749; non-rumor (training set): 7334; rumor (testing set): 3783; non-rumor (testing set): 5599
Twitter15 [77]	Users: 276 663; source tweets: 1490; threads: 331 612; non-rumors: 374; false rumors: 370; true rumors: 372; unverified rumors: 374; average time per tree: 1337 h; average posts per tree: 223
Twitter16 [77]	Users: 173 487; source tweets: 818; threads: 204 820; non-rumors: 205; false rumors: 205; true rumors: 205; unverified rumors: 203; average time per tree: 848 h; average posts per tree: 251
FakeNewsNet [73]	The dataset does not provide specific details like the previous dataset but encompasses a wide range of attributes, including news content, social context, and spatiotemporal information
Kaggle Fake News Dataset [78]	The Kaggle Fake News dataset provided for the competition consists of approximately 20 800 articles, with each article classified as either “fake” or “real.” The dataset is primarily designed to help participants develop ML models capable of identifying misleading news content.
CoAID [70]	News articles (Version 0.2): 1339; user engagements (Version 0.2): 111 128; news articles (Version 0.3): 1016; user engagements (Version 0.3): 1308; timeframe for COVID-19 information: 1 May 2020 to 1 September 2020
Covid-HeRA [71]	Misinformation class imbalance present; average time between tweet and reply: 11–12 h (varies by category); total unique words: 1221
ArCOV19 [72]	Tweets in claims subset: 9414; false: 18.6%; true: 19.4%; other: 61.9%; tweets with replies: 3161; tweets with retweets: 3810; verified claims: 138

summaries of literary materials [107]. The sophistication of LLMs, such as GPT-3 and GPT-4, poses a challenge to existing methods of fake news detection [103], [104], [105].

B. Detecting LLM-Generated Content

Several indicators exist to discern machine-generated text. One method involves spotting inconsistencies or errors, as LLMs might occasionally exhibit minor logical lapses or factual discrepancies. Another technique is to scrutinize both content and style. Owing to their training on extensive datasets, LLMs’ outputs might echo the inherent patterns and biases of those datasets. Unconventional or skewed viewpoints in the text might suggest machine generation. Analyzing text for contextual relevance and coherence is also telling. While LLMs are adept at crafting contextually appropriate and coherent narratives, they may falter in ensuring uniformity or grasping subtle contexts [102]. Moreover, they may struggle with tasks that require more complex reasoning or multistage reasoning/inference. Any sudden tonal or thematic deviations could be indicative of machine generation. However, it is pivotal to acknowledge the non-infallible nature of these markers, especially as LLMs evolve [108]. In the context of watermarking, the technique is designed to discern machine-generated content. The strategy embeds discernible patterns in the produced content, striving to maintain output quality and variety. Study by [109] proposition of watermarks for natural language has evolved for use in neural language model outputs [109]. A recent contribution in this domain for transformer-based models is the adversarial watermarking transformer (AWT) by [110]. However, a model-agnostic

approach was introduced by [111], allowing for watermarking across prevalent autoregressive language models. Nonetheless, the method presents challenges, necessitating vast data storage and optimized retrieval mechanisms to efficiently manage escalating text volumes [108]. As their sophistication grows, discerning machine-generated from human-authored content could become more intricate. Hence, a vigilant approach to source verification remains indispensable [106], [112], [113].

Multiple techniques exist for identifying machine-produced text. A few prevalent methodologies include as follows.

- 1) **Linguistic examination:** Texts crafted by machines might manifest distinct linguistic attributes that are not typical in human-composed content. Studies have formulated specific linguistic attributes and metrics, including n-gram occurrences, syntactical configurations, and semantic consistency, to differentiate between texts produced by humans and machines [114], [115]. In an experimental study conducted by [116], the objective was to distinguish between human-authored and machine-generated text. The dataset employed for this research was sourced from the William and Flora Hewlett Foundation, who initiated a 2012 Kaggle contest to formulate an auto-scoring algorithm tailored for student essays. This dataset encompassed 12,978 essays penned by students spanning grades 7 to 10 [116].
- 2) **Style and tone assessment:** Texts derived from machines might miss the subtleties and unique flair intrinsic to human-authored content. Evaluating the stylistic approach, emotional tone, and overall sentiment can

TABLE IV
SUMMARY OF AI AND ML METHODS FOR FAKE NEWS DETECTION

Dataset	Reference	Feature Extraction	Learning Method	Result in Accuracy
MediaEval	[79]	ELA images, text preprocessing	EfficientNetB0, RoBERTa	85.30%
	[14]	Textual (BERT), visual (VGG-19)	Multimodal integration of text and visual features, classification via neural network	Improved macro-F1 scores over baselines by 3.27% (MediaEval)
PHEME & PHEME updated version	[80]	Structural, user, content, temporal features (e.g., tweet volume, account ages, polarity, and temporal trends)	Random Forest	ROC-AUC of 36.52%
	[81]	Tweet embeddings (TE), user credibility information embeddings (UE), tweet feature embeddings (FE)	Multitask learning with LSTM, attention mechanism	48.3% accuracy, 41.8% macro F1
	[82]	Latent representations extracted using LSTM-based variational autoencoder (VAE)	LSTM-based VAE, multitask learning (integrates rumor detection, tracking, stance, and veracity classification)	Up to 26.9% improvement in macro-F1 scores over state-of-the-art techniques
Weibo	[14]	Textual (BERT), visual (VGG-19)	Multimodal integration of text and visual features, classification via neural network	Improved macro-F1 scores over baselines by 6.83% (Weibo)
	[64]	Textual (sentiment analysis), visual (preprocessing and segmentation)	Cultural algorithm with normative and situational knowledge	89.1%; outperforms state-of-the-art by approximately 9%
	[83]	word2vec, BERT, Bi-LSTM	Proposed a domain-adversarial and graph-attention neural network (DAGA-NN) model	90.20%
Twitter15 & Twitter16	[84]	Textual (RoBERTa encodings), graphical (graphical convolutional networks)	Hybrid model combining graphical convolutional networks and Bi-LSTM layers	High classification performance, improves over state-of-the-art methods (specific values not detailed)
	[65]	Text semantic attention for tweet text, propagation structure attention for user interactions	Multiview attention networks (MVAN) combining RNN and graph neural networks with attention mechanisms	Outperforms state-of-the-art models by 2.5% on average in accuracy
	[85]	Word embeddings, multihead attention for local context, heterogeneous graph features	Global-local attention network (GLAN) that integrates local semantic and global structural information	Twitter15: 90.5%, Twitter16: 90.2%
FakeNewsNet	[86]	Textual features from BERT, LSTM for sequence modeling	Integration of BERT and LSTM to enhance feature extraction and sequence learning	Subdataset: politifact & Gossipcop; politiFact: 88.75%, GossipCop: 84.10%
	[87]	Derived tweet attributes, URL attributes, title word count	ML classification using Naïve Bayes, SVM, logistic Regression, decision tree, random forest	Best model (random forest) achieved 98.6% accuracy, 95.4% recall, and 97.2% F1 score
COVID-19 datasets	[88]	TF-IDF, Ngrams	Modified LSTM	94.21%
	[89]	Count vectorization and TF-IDF	LR, SVC, NB	87%
	[63]	Bag-of-words model with uni-gram features extracted	Capsule network model	79%
Kaggle Fake News Dataset	[13]	Global text semantics (BERT), local text semantics (CNN using N-gram features)	Integration of BERT for global semantics and CNN for local semantics into a dense network	Achieved 1.48% improvement over state-of-the-art

assist in pinpointing machine-crafted outputs. Tools such as sentiment analysis, stylometric evaluation (evaluation based on elements such as sentence structure, word choice, and punctuation usage [117]), and scrutiny of a particular word or phrase utilization can facilitate this differentiation [106], [118], [119].

- 3) **Knowledge-based analysis:** Texts produced by machines may not consistently showcase current or accurate knowledge. Determining the machines' origins can be achieved by validating the factual precision and consistency within the text. Methods such as fact verification, cross-referencing against credible resources, or pinpointing logical discrepancies are effective in this endeavor [106], [114], [118].
- 4) **Adversarial testing:** The approach of adversarial examination encompasses creating particular challenges or assignments. It examines how a model reacts to inputs that may be specially crafted to exploit the model's weaknesses or cause it to make incorrect predictions. Such tests might contain queries demanding elementary logical reasoning, discerning humor, or participating in unstructured dialogues. This can be especially useful in the context of fake news detection, as it helps to ensure that the model can handle deceptive information and attempts to mislead it [96], [97].

It is imperative to acknowledge that these methodologies have inherent limitations and are not infallible. As machine models persistently advance, there is a continuous interplay between the refinement of detection strategies and the evolution of more intricate machine-generated content. Consequently, amalgamating diverse techniques and consistently enhancing detection mechanisms is vital [97], [120].

VIII. LLMs TO DETECT FAKE NEWS

LLMs are advanced AI systems designed to understand, generate, and manipulate human language on a large scale. These models are extensively trained on vast and diverse text datasets, enabling them to perform a wide array of language-related tasks with high proficiency. Examples of LLMs include generative pretrained transformer (GPT), bidirectional encoder representations from transformers (BERT), A Lite BERT (ALBERT), and Text-To-Text Transfer Transformer (T5) [121], [122], [123]. In the preceding sections, we discussed the nature of LLMs, their operational mechanisms, and their ability to produce text that resembles human writing. We also examined the features that enhance their performance and the methods used to detect machine-generated content. Now, we will explore the opposite perspective: using LLMs to detect fake news. LLMs, such as GPT and BERT, are initially trained on extensive corpora covering a wide range of linguistic structures and contexts [103], [104], [124], [125]. To adapt these models for detecting fake news, several critical steps are necessary. First, a carefully curated, labeled dataset containing both genuine and fake news articles is required, ensuring a balanced and diverse representation of sources and topics. After collecting this data, the pretrained LLM undergoes fine-tuning with this

dataset, allowing the model to adjust its internal parameters specifically for fake news detection. Following this calibration, it is crucial to rigorously evaluate the model's performance on unseen data to confirm its accuracy and effectiveness in identifying misinformation [126], [127], [128]. This raises an important question: can LLMs also generate fake news? If so, can they verify their own output for factual accuracy [96], [97], [106]?

A. Methodologies and Algorithms

This section will explore the computational techniques and strategies employed for fake news detection, delving into prominent methodologies like transfer learning, few-shot learning, LoRA fine-tuning, and retrieval-augmented generation (RAG).

1) *Transfer Learning:* Transfer learning is a methodology in ML where a model developed for a task is reused as the starting point for a model on a second task [129]. It is a popular approach in deep learning where pretrained models are used as the starting point on computer vision and NLM tasks given the vast computing and time resources required to develop neural network models on these problems and from the enormous jumps in skill that they provide on related problems [129], [130]. Transfer learning can be beneficial in the context of fake news detection. For instance, a model trained to understand the semantics and context of text in one domain can be transferred to identify and classify fake news in another domain. This can be especially beneficial when dealing with fake news, as obtaining labeled training data for such a task can be challenging and time-consuming. By leveraging the knowledge gained from a related task, transfer learning can potentially improve the performance of a model in identifying and classifying fake news [131].

2) *Few-Shot Learning:* Few-shot learning is a ML technique where the model is given a few demonstrations of the task at inference time as conditioning, but no weights are updated [103], [132]. This reduces the need for a large amount of task-specific data [133]. However, the results from this method have so far been worse than state-of-the-art fine-tuned models.

3) *LoRA Fine-Tuning:* LoRA fine-tuning is an efficient method to adapt pretrained LLMs to new tasks without extensive computational resources. LoRA focuses on updating a low-rank representation of the model's weights, significantly reducing the number of trainable parameters while maintaining performance [134]. This technique is particularly useful for resource-constrained environments where full fine-tuning is not feasible. In the context of fake news detection, LoRA fine-tuning allows for rapid adaptation of LLMs to identify misinformation with high accuracy, leveraging limited labeled data effectively.

4) *RAG:* RAG integrates a retrieval mechanism with a generation model to enhance the performance of language models on tasks requiring external knowledge [135]. In fake news detection, RAG can be used to retrieve relevant factual information from a database to support the verification of claims made in news articles. This approach not only improves the

accuracy of fake news detection but also provides context and justification for the model's decisions, making the process more transparent and explainable.

5) *Siamese Networks and Triplet Loss*: Siamese networks are a type of neural network architecture containing two or more identical subnetworks with the same configuration and share the same parameters. These networks are typically used in tasks that involve finding similarities or relationships between two comparable things. On the other hand, triplet loss is a loss function used in ML that is designed to learn discriminative features. It does this by comparing an anchor instance to a positive instance (same class as anchor) and a negative instance (different class from anchor). The goal of triplet loss is to ensure that the anchor is closer to the positive instance than it is to the negative instance by some margin [136], [137]. In the context of fake news detection, these techniques can be used to compare the features of a given piece of news (the anchor) to known examples of real news (positive instances) and fake news (negative instances). The model is trained to minimize the triplet loss, which effectively means it is learning to distinguish between real and fake news accurately [136].

6) *Transformers and Attention Mechanisms*: Transformers are a model architecture used in deep learning, introduced by Vaswani et al. [95], known for using self-attention mechanisms. These can be trained to recognize patterns in the language used in fake news articles. For example, the efficient transformer based multilevel attention (ETMA) framework has set a state-of-the-art benchmark for fake news detection, achieving high accuracy with scores of 93%–97% [138].

B. Case Studies and Examples

This section reviews pivotal studies on the application of transformer embeddings, sentence transformers, and the explainability of LLMs in fake news detection, highlighting their effectiveness and challenges.

1) *Use of Transformer Embeddings in Fake News Detection*: Several studies demonstrate the adaptability of transformer models in distinguishing real information from fake news and satire. For instance, a DistilBERT-based model designed to differentiate fake news from satire achieved notable efficiency [139]. Another study used the DT-transformer and CLIP ViT L/14 encoder for multimodal misinformation detection, showing the effectiveness of a hybrid approach [140].

2) *Sentence Transformers in Fake News Detection*: Research has explored the utility of sentence transformers in both monolingual and cross-lingual fake news detection, emphasizing the role of these models in overcoming language barriers [141], [142].

3) *LLM Explainability and Fake News*: LLMs are effective in content-based fake news detection tasks but lack interpretability. Efforts to make LLMs more explainable include using SHapley Additive exPlanations (SHAP) and the hierarchical step-by-step (HiSS) prompting method for news claim verification [143], [144].

IX. TECHNIQUES IN FAKE NEWS DETECTION

This new chapter focuses on advanced techniques and strategies for detecting and mitigating fake news, including document embeddings, transformer models, deep neural network ensembles, and social network immunization.

A. Document Embeddings in Fake News Detection

Document embeddings provide rich, contextual representations of textual data, helping ML models capture nuanced semantic relationships. Studies have used sentence transformers and variational autoencoders to improve feature extraction processes, achieving high accuracy [145], [146].

B. Transformers in Misinformation Detection

Transformers have significantly advanced the detection of misinformation due to their sophisticated NLM capabilities. They improve generalization across diverse datasets and adapt to new and evolving data, essential for the dynamic landscape of news dissemination [147], [148].

C. Deep Neural Network Ensembles for Fake News Detection

Deep neural network ensembles (DNNs) improve performance by integrating textual and social context features, offering better accuracy and robustness than single models. The DANES framework, for example, combines multiple neural network branches to enhance model performance [149], [150].

D. Social Network Immunization Against Harmful Speech

Network immunization involves strategic interventions targeting key nodes within social networks to prevent the spread of harmful content. Methods like the Sparse Shield focus on immunizing influential nodes to minimize the spread of misinformation [151], [152].

E. Real-Time Mitigation Using Tree Algorithms in Social Media

Algorithms like time-aware crowd signals (TCS) and minimum-cost weighted directed spanning tree (MCWDST) optimize network interactions to minimize misinformation spread [149], [153].

F. Integrating Community Detection for Network Immunization in Fake News Detection

Integrating community detection algorithms into network immunization frameworks offers a strategic enhancement for combating fake news. By targeting specific clusters where misinformation is prevalent, interventions can be applied more effectively [154], [155].

G. Distributed Systems in Community-Based Misinformation Detection

The ContCommRTD system utilizes geolocated tweets to detect misinformation in disaster-related social media posts, forming communities based on content similarity and geographical proximity [156]. This method enhances real-time monitoring and management of misinformation.

X. FAKE-NEWS DETECTION BEFORE VERSUS AFTER LLMs

To summarize the aforementioned sections, a discernible distinction between the pre-LLM and LLM period regarding fake news detection can be outlined. Primarily, this distinction is evident in the detection methodologies employed. In the early stages of fake news detection, methods largely depended on rule-based systems, where manually designed rules or heuristics were applied. As technology progressed, there was a shift towards leveraging traditional ML techniques. Methods such as SVM, decision trees, and random forests became popular, especially when paired with meticulously engineered features. Upon closer investigation of artificial intelligence, early deep learning models, including CNN and RNN, were adopted to detect fake news in textual data. The transition to the era of LLM marked a significant enhancement in the sophistication of these methods. The emergence of transformer-based models, like BERT, revolutionized the field. They possess a profound ability to grasp context, enabling them to be fine-tuned on specialized fake news datasets. Furthermore, the capabilities of LLM have expanded the horizons of fake news detection through few-shot or even zero-shot learning, where the model is primed with minimal examples and is expected to generalize the task regarding the dataset used.

In the following paragraph, we will compare various elements of detecting fake news in both the pre-LLM and LLM period. These elements encompass: Section X-A: timeliness and scalability, Section X-B: resource intensiveness, Section X-C: adaptability and generalization, Section X-D: false positives and negatives, and Section X-E: bias and fairness.

A. Timeliness and Scalability

Traditional text detection and analysis methods, such as rule-based systems and earlier machine-learning models, often exhibit limitations in both timeliness and scalability. These methods typically demand manual interventions like rule updating or periodic retraining, particularly when adapting to new topics or evolving news contexts, making their response to dynamic data less immediate [10], [157], [158]. On the other hand, LLM is fundamentally different. They come pretrained on vast datasets, allowing them to process massive amounts of text data efficiently and ensuring rapid adaptability to new information [159], [160], [161]. Unlike their traditional counterparts, LLMs do not necessitate rule-based updates for each new topic, granting them an edge in real-time text analysis. This distinction becomes especially pronounced in news cycles, characterized by rapid evolution, diverse sources, and the demand for immediate responses. Thus, for applications ranging from live event

coverage to real-time sentiment analysis, LLMs appear to offer a more timely and scalable solution [159], [160], [161].

B. Resource Intensiveness

Traditional methodologies predominantly rely on extensive human input and temporal investments in the context of fake news detection. These conventional approaches necessitate the involvement of interdisciplinary teams comprising fact-checkers, journalists, and domain experts for rigorous information verification [32], [162], [163]. The hands-on nature of these procedures is resource-intensive, presenting challenges in staying up-to-date with the rapid proliferation of digital news content. Moreover, there's a considerable commitment of resources towards training and equipping novice fact-checkers with specialized knowledge. In contrast, contemporary LLMs exert significant computational demands. Training these sophisticated models entails using cutting-edge GPUs or TPUs, massive data storage capacities to accommodate extensive datasets and notable energy consumption, raising ecological concerns. Nonetheless, post-training, LLMs can maximize, streamline and automate the fake news identification process, optimizing human resource utilization during real-time evaluations [159], [164].

C. Adaptability and Generalization

In the domain of fake news detection, traditional techniques have predominantly leaned on engineered features. While these techniques demonstrate effectiveness within specific domains, their adaptability to the dynamic nature of misinformation remains a challenge [10], [165], [166], [167]. Such approaches often face issues with generalization, potentially leading to overfitting or ineffective knowledge transfer across varied contexts or languages [10], [165], [166], [167]. Conversely, LLM introduces a versatile avenue. With their ability to learn from extensive datasets, these models reduce the dependence on manual feature design [168], [169], [170]. They possess an intrinsic capability to recognize diverse linguistic subtleties, which provides them with increased generalization. However, similar to conventional techniques, LLMs have their limitations. They can exhibit overfitting without appropriate fine-tuning and may even amplify biases from their training data. Given the continuous evolution of fake news tactics, no single model can guarantee sustained effectiveness [168], [169], [170]. This highlights the significance of constant adaptability. A pragmatic strategy might combine the strengths of both traditional and LLM-based methods, targeting a robust, unbiased, and flexible fake news detection framework.

D. False Positives and Negatives

The propensity for false positives in conventional ML methodologies is often attributed to their dependence on manually-defined features and susceptibility to overfitting, resulting in erroneous classifications [8], [10], [171], [172]. Conversely, the inability to generalize beyond established features and dataset biases often leads to false negatives. Modern

expansive language models, though proficient in textual understanding, are not immune to false positives, a consequence of biases within training datasets, the absence of an anchor in real-world veracity, and a heightened sensitivity to nuanced linguistic cues [168], [169], [170]. Furthermore, these models' capacity for discernment is tested when faced with artfully constructed fake news that mirrors authentic content and challenges in comprehending the broader deceptive context. The ever-adapting nature of deceptive strategies and ingrained training data biases exacerbates these issues. An amalgamation of advancing AI strategies and human discernment is imperative for a robust approach to fake news detection.

E. Bias and Fairness

Bias and fairness in ML are critical considerations, especially in sensitive applications like fake news detection. The cutting-edge LLMs and the more traditional detection systems carry inherent challenges in this context. LLMs draw their knowledge from massive volumes of Internet data. If this foundational data is biased, the LLM can potentially echo and even magnify those biases [117], [173]. For instance, an undue negative representation of a specific group in training data can skew the LLM's responses towards that bias. Furthermore, if a particular viewpoint dominates its training data, the LLM might misleadingly treat it as the mainstream perspective biased, and the LLM can potentially echo and even magnify those biases [117], [173]. On the other hand, bias in ML can occur due to various reasons, including measurement bias, where certain features are chosen, utilized, and measured in a particular way [174], [175], [176]. For instance, training data selection for fake news detection could be biased, leading to unfair outcomes. Moreover, ML algorithms can inadvertently amplify existing biases in the data. For example, suppose the training data for fake news detection is predominantly sourced from certain types of news outlets or focuses on specific topics. In that case, the resulting model may unfairly target or favor certain groups or types of news [174], [175], [176]. Addressing these biases is crucial to ensure fairness in ML applications. Various strategies have been proposed, including the use of calibration algorithms like RBA (reducing bias amplification) to ensure that model predictions follow the same distribution as the training data [174], [175], [176]. However, it is essential to note that fairness in ML is a complex issue that extends beyond the technical aspects of algorithm design and implementation. It also involves ethical and societal considerations, such as the potential for discriminatory behavior towards certain groups or populations [174], [175], [176]. Therefore, ongoing research and dialogue are necessary to navigate these challenges and ensure the responsible use of ML in applications like fake news detection.

XI. CONCLUSION

The advent of LLMs has exacerbated the complexity of fake news detection. Fake, inherently multidimensional news spans from blatant factual inaccuracies to subtle misleading implications. With the sophistication of LLMs, this issue is further compounded, as they can produce content that

complicates detection methodologies, especially when considering transferability across diverse languages and cultural contexts. Distinct challenges also arise from LLMs' intricate handling of sarcasm and humor. Even though LLMs necessitate vast datasets for training, the acquisition of datasets, specifically annotated for fake news, is non-trivial. The swift evolution of LLM capabilities establishes an intricate detection dynamic compounded by the latent risk of model-induced biases.

Despite these challenges, LLMs offer notable opportunities. The same architectures capable of generating misinformation can be strategically adapted for its identification. Integrating LLMs with advanced image or video analytics makes a multi-modal approach to misinformation detection feasible. Such synergy facilitates real-time misinformation analysis, potentially mitigating its widespread propagation. A notable advantage emerges from community-driven initiatives, fostering robust, diverse detection solutions. Moreover, the educational potential of LLMs can be harnessed to design tools to enhance public discernment of misinformation.

The integration of human expertise and AI capabilities is crucial for future developments. A prospective direction involves the design of LLMs that engage in incremental learning, adapting to the dynamic nature of misinformation. The profound influence of LLMs necessitates comprehensive discourse on ethical considerations and potential regulatory frameworks. A focus on model transparency and decision explainability will be indispensable. A holistic detection approach, recognizing the ubiquity of misinformation across platforms, underscores the importance of multiplatform collaboration. Additionally, the broad spectrum of stakeholders, including journalists and policymakers, warrants targeted education about LLM capabilities and constraints. By leveraging strategic collaboration, continuous innovation, and stakeholder awareness, the challenges induced by LLMs can be strategically transformed into instrumental opportunities.

In the digital age, the distinction between authentic information and fake news has become increasingly nebulous, posing significant challenges for readers, policymakers, and technologists alike. The emergence of fake news, especially after the 2016 U.S. elections, underscored the profound implications disinformation can have on societal dynamics and democratic processes. The journey through history, from the rudimentary origins of fake news to the complexities introduced by LLMs, paints a picture of an ever-evolving landscape.

Traditional methods of fake news detection, based on standard ML approaches, provided a foundational understanding and initial remedies. However, with their immense capabilities, the advent of LLMs has simultaneously opened doors to incredible advancements and potential misuse. The line between human-generated content and machine-generated narratives has blurred, demanding more sophisticated detection mechanisms.

In light of the diversified data sources available today, solely relying on text-based detection mechanisms is both myopic and limiting. The nuances in detecting fake news necessitate a comprehensive approach that embraces a wider array

of data types. Incorporating multimodal techniques, which synergistically combine various data types such as text, images and social interactions can drastically improve the accuracy and robustness of detection systems. This multimodal perspective not only enhances the discernment capabilities but also addresses the multifaceted nature of disinformation, ensuring a more holistic safeguarding of authentic information. Hence, as we further our efforts in combating fake news, the integration of multimodal methods will be indispensable in navigating the intricacies of the modern digital landscape.

The comparison between pre-LLM and LLM-era techniques provides valuable insights into our progress and the monumental challenges that still lie ahead. We must continually reassess our strategies and tools as we navigate this intricate domain. A balanced combination of technological innovation, media literacy education, and regulatory frameworks will be essential in ensuring a trustworthy digital information ecosystem.

In essence, our exploration into fake news in the context of LLMs emphasizes the need for vigilance, adaptability, and collaboration across disciplines. The fight against fake news isn't just a technological challenge; it's a societal one, requiring a comprehensive, multifaceted and multilevel approach to safeguard the integrity of information in our interconnected world.

XII. FUTURE DIRECTIONS FOR LLMs IN FAKE NEWS DETECTION

Fake news is a widespread and intricate problem that affects many areas of society, from politics to academia. In our era of extensive data, the spread of misinformation has significant consequences. While there have been many advances in detecting fake news using ML and integrating LLMs, we now face the additional challenge of LLMs creating fake news. This highlights the need for a more detailed and specialized approach to detecting fake news.

A. Segmenting Fake News Detection

To improve the effectiveness of fake news detection, efforts should be divided into more focused and specialized areas.

- 1) **Domain-specific detection:** Focusing on specific domains such as political news, health misinformation, or academic fraud can lead to more accurate and effective detection systems. Each domain has unique traits and challenges, requiring tailored approaches.
- 2) **Type-specific detection:** Differentiating between types of misinformation, such as biased reporting, false claims, or satire, can improve detection accuracy. Specialized models can be developed to address the nuances of each type.
- 3) **Early-stage detection:** Prioritizing the early detection of fake news can help prevent its spread before it becomes widespread. Developing systems that identify and flag potential misinformation as it appears can be a proactive way to combat fake news.

B. Collaborative Efforts

The complexity of fake news detection calls for a multidisciplinary approach. It is not just a task for computer scientists but requires collaboration with experts from various fields.

- 1) **Interdisciplinary collaboration:** Cooperation between computer scientists, sociologists, psychologists, journalists, and policymakers is crucial. Each field brings valuable insights and expertise, leading to a more comprehensive approach to detecting and preventing misinformation.
- 2) **Ethical frameworks:** Establishing ethical guidelines is essential to address data privacy concerns and ensure the responsible use of AI technologies. These frameworks should promote transparency, accountability, and fairness in the development and use of detection systems.
- 3) **Community involvement:** Engaging the wider community, including researchers, developers, and the public, can help create diverse and resilient solutions. Public awareness and media literacy initiatives can empower individuals to critically evaluate information and recognize misinformation.

C. Enhancing Technology and Capabilities

Although we have made significant progress, there is still much work to be done to develop detection systems that are advanced and reliable enough to meet our needs.

- 1) **Incremental learning and adaptability:** Developing LLMs that can continuously learn and adapt to new patterns of misinformation is crucial. Incremental learning capabilities will allow models to stay up-to-date with evolving tactics used by those spreading fake news.
- 2) **Multimodal integration:** Combining LLMs with advanced analytics for images, videos, and social media interactions will enhance the robustness and comprehensiveness of fake news detection systems. Multimodal approaches can address the diverse and complex nature of misinformation across various platforms.
- 3) **Real-time misinformation analysis:** Creating systems capable of real-time analysis and detection of misinformation can help limit its rapid spread. Using the speed and efficiency of LLMs along with real-time data from multiple sources will be key in this effort.
- 4) **Explainability and transparency:** Improving the explainability of LLMs' decisions and making their processes more transparent is essential. Users need to understand how conclusions are reached in order to trust and effectively use these systems.

In summary, the future of LLMs in fake news detection lies in their ability to continuously adapt, integrate multimodal data, and collaborate with human expertise. As technology evolves, a balanced approach combining innovation, ethical considerations, education, and collaboration will be vital in combating the widespread challenge of fake news. The battle against misinformation is not just a technological issue but a societal one, requiring a comprehensive, multifaceted approach to ensure the integrity of information in our interconnected world.

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APPENDIX A

SCOPUS ADVANCED SEARCH QUERY

The following advanced search query was used in Scopus:

```
(TITLE-ABS-KEY("fake news detection" OR
"misinformation identification" OR "
disinformation detection" OR "fake
news propagation" OR "content
authenticity" OR "fact-checking
algorithms" OR "automated fact-
checking" OR "real-time detection" OR
"social media analysis") AND
TITLE-ABS-KEY("large language model*" OR
"natural language processing" OR "NLP"
OR "deep learning" OR "machine
learning" OR "transformer model*" OR
"BERT" OR "GPT-3" OR "GPT-4" OR "neural
network*" OR "text classification" OR
"sentiment analysis" OR
"computational linguistics" OR "data
mining" OR "information retrieval" OR
"AI ethics" OR "trustworthy AI" OR
"cross-lingual fake news detection" OR
"multimodal fake news detection" OR
"adversarial attacks on NLP models" OR
"model interpretability"))
```

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